20 Logistic Regression

Logistic regression is a special type of nonlinear regression developed for a binary response variable (Y), for example:

- Disease status: yes/no
- Treatment response: responder/non-responder
- Smoking: current smoker/not
- Mortality: dead/alive

A common use for logistic regression is to fit a dose-response curve, where mortality (or survival) of individuals (the response) is plotted agins the concentration of a adrug, toxin, or other chemical.

If we tried linear regression on a 0/1 outcome, predicted values could fall outside ($[0, 1]$ and error assumptions would be violated. Logistic regression models the **probability** in a way that stays in $[0, 1]$.

20.1 The logistic model

Let $Y \in \{0, 1\}$ be the outcome and X be a predictor (or a set of predictors). Define:

$$p(x) = P(Y = 1 \mid X = x)$$

$$p(x) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Logistic regression uses the **logit link**:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

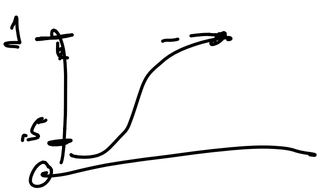
Equivalently, the predicted probability is:

$$\frac{1}{1+exp(x)}$$

$$p(x) = \frac{1}{1 + \exp\left(-(\beta_0 + \beta_1 x)\right)}$$

sigmoid function of $\beta_0 + \beta_1 X$

This creates an S-shaped curve mapping real numbers to probabilities.



$X = 0$

$Y = 1$

Example:

$\exp(\log(x)) = X$

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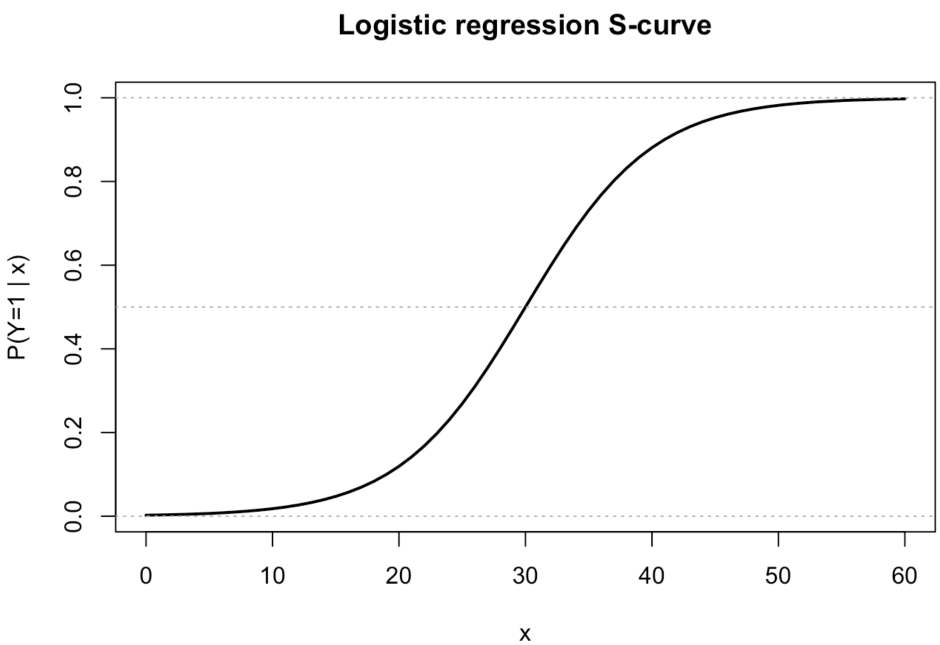
$\frac{p}{1-p} = e^{\beta_0 + \beta_1 X}$

$= e^{\beta_0} \cdot e^{\beta_1 X}$

```
beta0 <- -6
beta1 <- 0.2
x <- seq(0, 60, by = 1)
p <- 1/(1 + exp(-(beta0 + beta1*x)))

plot(x, p, type = "l", lwd = 2,
     xlab = "x", ylab = "P(Y=1 | x)",
     main = "Logistic regression S-curve")
```

```r
abline(h = c(0, 0.5, 1), lty = 3, col = "gray70")
```



20.1.1 Log-odds interpretation

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$$\beta_1 = \text{change in log-odds of } Y = 1 \text{ for a 1-unit increase in } x$$

20.1.2 Odds ratio interpretation

Exponentiating gives an **odds ratio (OR)**:

$$\exp(\beta_1) = \text{multiplicative change in odds for a 1-unit increase in } x$$

- If $\exp(\beta_1) = 1$: no association
- If $\exp(\beta_1) > 1$: higher odds as x increases
- If $\exp(\beta_1) < 1$: lower odds as x increases

Example: If $\exp(\beta_1) = 1.50$, each 1-unit increase in x is associated with **50% higher odds** of $Y = 1$, holding other variables constant.

20.2 Multiple logistic regression

With multiple predictors:

$$\text{logit}(p) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k \; + \beta_{k+1} x_k^2$$

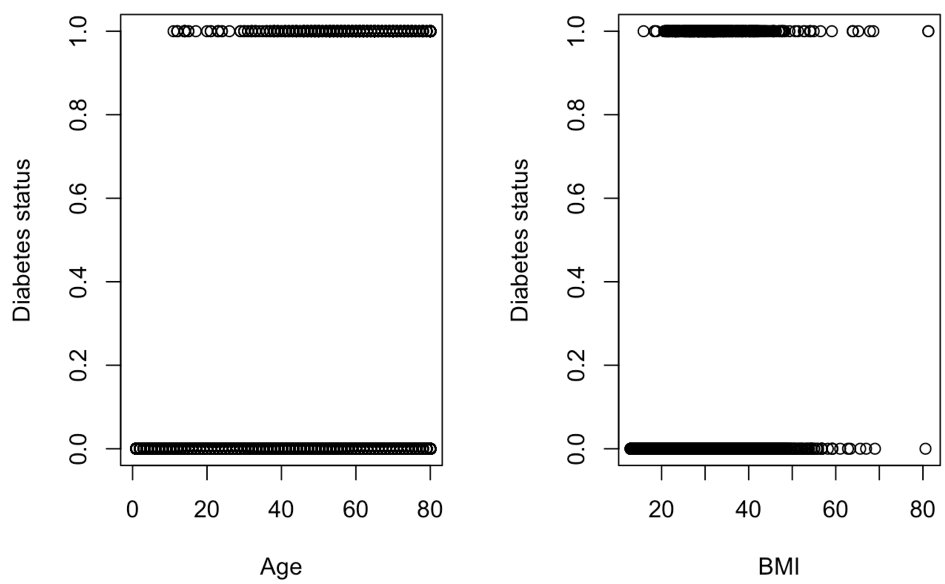
Interpretation of $\exp(\beta_j)$: adjusted odds ratio for x_j, **holding all other predictors fixed**.

This is often used to adjust for confounding in observational studies.

For example, we might model the probability of developing diabetes as a function of BMI and age

```
library("dplyr")
library(NHANES)
NHANES<-NHANES %>%
  mutate(has_diabetes=as.numeric(Diabetes == "Yes"))

par(mfrow=c(1,2))
    plot(NHANES$Age,NHANES$has_diabetes, xlab="Age",ylab="Diabe
    plot(NHANES$BMI,NHANES$has_diabetes, xlab="BMI",ylab="Diabe
```



We see that the probability that a subject has diabetes tends to increase as both a function of age and BMI.

Which variable is more important: Age or BMI? We can use a multiple logistic regression model to model the probability of diabetes as a function of both predictors

```
logreg<-glm(has_diabetes~BMI+Age, family="binomial",data=NHANES
summary(logreg)
```

```

Call:
glm(formula = has_diabetes ~ BMI + Age, family = "binomial",
    data = NHANES)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -8.080292   0.244447  -33.05   <2e-16 ***
BMI          0.094325   0.005519   17.09   <2e-16 ***
```

```
Age          0.057278   0.002487   23.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5263.8  on 9628  degrees of freedom
Residual deviance: 4146.0  on 9626  degrees of freedom
  (371 observations deleted due to missingness)
AIC: 4152

Number of Fisher Scoring iterations: 7
```

Both predictors seem to be important. To interpret the findings, we might consider a visual display of predicted probabilities as

```
library(modelr)
fake_grid<- data_grid(
  NHANES,
  Age =seq_range(Age,100),
  BMI=seq_range(BMI,100)
)

y_hats<- fake_grid %>%
  mutate(y_hat=predict(logreg,newdata=.,type="response"))
head(y_hats,1)
```

```
# A tibble: 1 × 3
    Age   BMI   y_hat
  <dbl> <dbl>   <dbl>
1     0  12.9 0.00104
```

20.3 Interpreting coefficients

The predicted probability from the model is given by:

$$\pi_i = P(y_i = 1 \mid Age_i, BMI_i) = \frac{e^{\beta_0+\beta_1 Age_i+\beta_2 BMI_i}}{1+e^{\beta_0+\beta_1 Age_i+\beta_2 BMI_i}}$$

Let's consider a hypothetical 0 year old with a BMI of 12.9. Their predicted probability of having diabetes would be calculated as a function of the regression coefficients:

```
linear_component<-c(1,12.9,0)%*%coef(logreg)
exp(linear_component)/(1+exp(linear_component))
```

```
            [,1]
[1,] 0.001044164
```

The predicted probability is very small, what about a 60 year old with a BMI of 25?

```
linear_component<-c(1,25,60)%*%coef(logreg)
exp(linear_component)/(1+exp(linear_component))
```

```
           [,1]
[1,] 0.09233095
```

The predicted probability is now 9.2%

20.4 Categorical predictors

Suppose sex is coded as:

- Female = 0 (reference)
- Male = 1

Then:

$$\exp(\beta_{\text{male}}) = \frac{\text{odds of } Y = 1 \text{ for males}}{\text{odds of } Y = 1 \text{ for females}}$$

For predictors with more than two categories, use indicator (dummy) variables with a reference group.

Some notes for regression in general.

- If x^2 is important to consider, you should also include x

  so: $y = \beta_0 + \beta_1 x + \beta_2 x^2$

  not just

  $\beta_0 + \beta_2 x^2$

  for ease of interpretation

- Interactions

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

models interactions.